DSA03-Natural Language Processing

List of Lab Program

20. Implement a basic information retrieval system using TF-IDF (Term Frequency-Inverse Document Frequency) for document ranking using python.

Code:

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
documents = [
    "Python is used for machine learning",
    "Java is used for application development",
    "Machine learning uses artificial intelligence"
]
query = ["machine learning"]
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(documents + query)
similarity = cosine_similarity(
    tfidf_matrix[-1], tfidf_matrix[:-1]
)[0]
print("Document Ranking:")
ranking = similarity.argsort()[::-1]
for i in ranking:
    print(f"Document {i + 1}: {documents[i]}")
    print(f"Similarity Score: {similarity[i]:.4f}")
```

Output:

```
Document Ranking:
Document 1: Python is used for machine learning
Similarity Score: 0.4706
Document 3: Machine learning uses artificial intelligence
Similarity Score: 0.4622
Document 2: Java is used for application development
Similarity Score: 0.0000
```